

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1272

North Blacksburg 138 kV
September 03, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1272. The Interconnection Customer proposes a 163.9 MW wind facility interconnecting at the North Blacksburg 138 kV substation in the Kentucky area. The request was received on August 24, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 480 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,950,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Norwich Substation to Barrett 500.0 kV, the Toledo Bend to Wharton 230.0 kV Line loads to 111.1% of its 702 MVA rating. Project DFAX: 8.7%.

The Valemount to Pence 345.0 kV is overloaded to 109.7% of its 1,288 MVA emergency rating for the loss of the Gavin Power Llc to Sporn 230.0 kV. The project contributes 7.6% DFAX (12.5 MW) to this overload.

The Valemount to Tristate 345.0 kV is overloaded to 104.8% of its 1,501 MVA emergency rating for the loss of the Sporn Substation to Gavin Power Llc 230.0 kV. The project contributes 7.9% DFAX (12.9 MW) to this overload.

For the loss of E Methuen to Lonetree Substation 500.0 kV, the Arpgen to N Blacksburg 138.0 kV loads to 103.6% of its 257 MVA rating. Project DFAX: 100.0%.

Arpgen to Sierra Pine Substation 138.0 kV: 101.9% loading (248 MVA) following the outage of the Killen to Moffat 500.0 kV; DFAX = 100.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Toledo Bend to Wharton 230.0 kV Line	Rebuild line with higher-capacity conductor	\$14,433,000	3% (\$438,000)
Valemount to Pence 345.0 kV	Reconductor line	\$10,928,000	1% (\$150,000)
Valemount to Tristate 345.0 kV	Reconductor line	\$31,118,000	1% (\$408,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Arpgen to N Blacksburg 138.0 kV	Reconductor line	\$5,956,000	100% (\$5,956,000)
Arpgen to Sierra Pine 138.0 kV	Reconductor line	\$3,998,000	100% (\$3,998,000)
	Total allocated to project		\$10,950,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.